

Telco Customer Churn Dataset - Complete Documentation

Dataset Overview

This dataset contains customer information from a telecommunications company, focusing on customer retention and churn analysis. It includes 21 features describing customer demographics, services, account information, and the target variable indicating whether a customer has churned.

Business Context

This is a **customer churn prediction** dataset for a telecommunications provider offering:

- Phone services (landline)
- Internet services (DSL, Fiber optic)
- Various add-on services (security, backup, streaming, etc.)
- Different contract terms and payment methods

The primary business objective is to identify customers at risk of churning to implement retention strategies.

Column Documentation

Customer Identification

- **customerID**: Unique identifier for each customer
 - Data Type: String
 - Format: 4 digits followed by hyphen and 5 letters (e.g., "7590-VHVEG")
 - Business Use: Customer tracking and record linking

Demographics

- **gender**: Customer's gender
 - Values: Male, Female
 - Business Use: Demographic segmentation and targeted marketing
- **SeniorCitizen**: Whether customer is a senior citizen
 - Values: 0 (No), 1 (Yes)
 - Note: Numeric encoding instead of Yes/No like other binary fields
 - Business Use: Age-based service offerings and pricing strategies
- **Partner**: Whether customer has a partner/spouse
 - Values: Yes, No
 - Business Use: Family plan opportunities and stability indicators

- **Dependents:** Whether customer has dependents
 - Values: Yes, No
 - Business Use: Family service bundles and churn risk assessment

Account Information

- **tenure:** Number of months customer has been with company
 - Data Type: Integer (0-72 months)
 - Range: 0-72 months observed in dataset
 - Business Use: Customer lifetime value, loyalty segmentation

Service Offerings

Core Services

- **PhoneService:** Whether customer has phone service
 - Values: Yes, No
 - Business Use: Service bundle analysis
- **MultipleLines:** Multiple phone lines status
 - Values: Yes, No, "No phone service"
 - Dependency: Only applicable when PhoneService = Yes
 - Business Use: Upselling opportunities
- **InternetService:** Type of internet service
 - Values: DSL, "Fiber optic", No
 - Business Use: Service tier analysis, infrastructure planning

Add-on Services (Internet-dependent)

The following services are only available with internet service:

- **OnlineSecurity:** Online security service
 - Values: Yes, No, "No internet service"
 - Business Use: Security service adoption, revenue per customer
- **OnlineBackup:** Online backup service
 - Values: Yes, No, "No internet service"
 - Business Use: Data service revenue tracking
- **DeviceProtection:** Device protection service
 - Values: Yes, No, "No internet service"
 - Business Use: Insurance revenue, customer care costs

- **TechSupport:** Technical support service
 - Values: Yes, No, "No internet service"
 - Business Use: Support cost analysis, service quality
- **StreamingTV:** TV streaming service
 - Values: Yes, No, "No internet service"
 - Business Use: Entertainment service adoption
- **StreamingMovies:** Movie streaming service
 - Values: Yes, No, "No internet service"
 - Business Use: Content service revenue

Contract & Billing

- **Contract:** Contract term length
 - Values: "Month-to-month", "One year", "Two year"
 - Business Use: Customer commitment levels, churn risk assessment
- **PaperlessBilling:** Whether customer uses paperless billing
 - Values: Yes, No
 - Business Use: Cost reduction initiatives, customer preferences
- **PaymentMethod:** How customer pays bills
 - Values: "Electronic check", "Mailed check", "Bank transfer (automatic)", "Credit card (automatic)"
 - Business Use: Payment processing costs, automation levels

Financial Information

- **MonthlyCharges:** Monthly bill amount
 - Data Type: Numeric (currency)
 - Range: Varies based on services
 - Business Use: Revenue analysis, pricing optimization
- **TotalCharges:** Total amount charged over tenure
 - Data Type: Numeric (currency)
 - Note: Contains some blank values for new customers (tenure = 0)
 - Calculation: Should approximate $\text{MonthlyCharges} \times \text{tenure}$
 - Business Use: Customer lifetime value, revenue forecasting

Target Variable

- **Churn:** Whether customer has left the company
 - Values: Yes (churned), No (active)

- Business Use: Primary prediction target for retention models

Key Performance Indicators (KPIs)

Customer Metrics

1. **Churn Rate:** Percentage of customers who cancelled service
 - Formula: $(\text{Churned Customers} / \text{Total Customers}) \times 100$
 - Industry Benchmark: 15-25% annually for telecom
2. **Customer Lifetime Value (CLV):** Expected revenue from customer
 - Formula: $(\text{Average Monthly Revenue} \times \text{Gross Margin \%}) / \text{Monthly Churn Rate}$
 - Use: Investment decisions in customer acquisition/retention
3. **Average Revenue Per User (ARPU):** Monthly revenue per customer
 - Formula: $\text{Total Monthly Revenue} / \text{Total Active Customers}$
 - Use: Pricing strategy and market positioning

Retention Metrics

4. **Customer Tenure Distribution:** How long customers stay
 - Segments: 0-12 months (new), 13-36 months (growing), 37+ months (mature)
 - Use: Identify critical retention periods
5. **Contract Mix:** Distribution of contract types
 - Use: Revenue predictability and churn risk assessment
6. **Service Penetration Rates:** Adoption of each add-on service
 - Formula: $(\text{Customers with Service} / \text{Total Customers}) \times 100$
 - Use: Upselling opportunities and bundle effectiveness

Financial KPIs

7. **Monthly Recurring Revenue (MRR):** Predictable monthly income
 - Use: Business health and growth tracking
8. **Revenue per Service:** Income contribution by service type
 - Use: Service profitability analysis
9. **Payment Method Distribution:** Customer payment preferences
 - Use: Processing cost optimization

Operational KPIs

10. **Service Bundle Penetration:** Multiple services per customer
 - Use: Cross-selling effectiveness

11. **Digital Adoption Rate:** Paperless billing percentage
 - Use: Operational cost reduction tracking
12. **Senior Citizen Retention:** Churn rates for senior customers
 - Use: Specialized retention programs

Business Insights Framework

Churn Risk Factors

- **High Risk:** Month-to-month contracts, electronic check payments, newer customers (< 12 months)
- **Medium Risk:** Single service customers, no add-ons, high monthly charges relative to services
- **Low Risk:** Long-term contracts, automatic payments, multiple services, longer tenure

Revenue Optimization

- **Upselling Targets:** Internet customers without add-on services
- **Contract Conversion:** Month-to-month to annual contracts
- **Payment Automation:** Manual payment customers to automatic methods

Customer Segmentation

1. **High-Value Loyalists:** Long tenure + high ARPU + low churn risk
2. **Growth Prospects:** Medium tenure + room for service expansion
3. **At-Risk Churners:** Short tenure + minimal services + month-to-month
4. **Price Sensitive:** High churn correlation with pricing changes

Data Quality Notes

- **Missing Values:** TotalCharges has blank entries for some new customers
- **Encoding Inconsistency:** SeniorCitizen uses 0/1 while other binaries use Yes/No
- **Logical Dependencies:** Add-on services depend on internet service availability
- **Calculation Verification:** TotalCharges should align with MonthlyCharges × tenure

Recommended Analysis Approaches

1. **Cohort Analysis:** Track customer behavior over time segments
2. **Survival Analysis:** Time-to-churn modeling
3. **RFM Analysis:** Recency, Frequency, Monetary customer segmentation
4. **Price Sensitivity Analysis:** Churn correlation with pricing tiers
5. **Service Bundle Analysis:** Optimal service combinations for retention